

STA-1 (Q9NAD6): The Canonical *Caenorhabditis elegans* STAT — A Tyrosine-Phosphorylation-Dependent, Repressive Transcription Factor Governing Dauer Formation and Antiviral Defense

Summary

STA-1 (gene *sta-1*, ORF *Y51H4A.17*, UniProt **Q9NAD6**) encodes the canonical **Signal Transducer and Activator of Transcription (STAT)** protein of the nematode *Caenorhabditis elegans*. It is a 706-amino-acid, multi-domain transcription factor belonging to the STAT family (InterPro IPR001217), carrying the hallmark STAT architecture: a coiled-coil domain, a p53-like DNA-binding domain (IPR008967), a STAT linker (IPR048988), and a C-terminal SH2 domain (IPR000980; residues ~477–574). Its primary molecular function is **tyrosine-phosphorylation-dependent, sequence-specific DNA binding** at a phylogenetically conserved STAT (GAS-like) *cis*-element, coupled to **transcriptional regulation**. STA-1 shuttles between the cytoplasm and nucleus, carrying out its function inside the cell as a signal-responsive transcription factor.

The defining and most surprising feature of STA-1 biology is that, in contrast to the activating mammalian STATs that drive interferon and cytokine responses, STA-1 functions predominantly as a **transcriptional REPRESSOR**. This repressive logic is expressed in two well-characterized physiological contexts. First, in a set of head and amphid neurons, STA-1 **represses the dauer developmental decision**, acting non-canonically alongside a specific branch (DAF-1/DAF-8, independent of DAF-7/DAF-4/DAF-14) of the DAF-7/TGF- β signaling pathway, with its own expression held in check by a DAF-3-dependent negative-feedback loop. Second, in the intestine, STA-1 acts **cell-intrinsically as a negative regulator of the antiviral and anti-pathogen transcriptional program**; loss of STA-1 makes animals *more* resistant to the natural Orsay virus, revealing that STA-1 normally suppresses a constitutive defense state.

Mechanistically, STA-1 operates **without a JAK kinase and without an interferon system**, neither of which exists in *C. elegans*. Its activation nevertheless requires a phosphorylatable tyrosine, implying an alternative (non-JAK) tyrosine kinase; the ACK1/TNK2-family tyrosine kinase **SID-3** has been identified as a genetic partner in the antiviral transcriptional network. Upon viral infection, STA-1 is removed from the nucleus of infected intestinal cells and forms **cytoplasmic puncta that interact with the RIG-I-like RNA viral sensor DRH-1**, a spatial switch that de-represses defense genes. Structural predictions place STA-1 closest to mammalian **STAT5**, which itself has documented immune-repressive roles. Importantly, STA-1 must be distinguished from its paralog **STA-2**, a separate STAT-like protein that governs epidermal antimicrobial-peptide responses — a frequent source of literature confusion addressed explicitly below.

Gene/Protein Identity Verification

Before presenting findings, the target identity was confirmed against the UniProt record and the primary literature:

Attribute	Value	Source
UniProt accession	Q9NAD6	UniProt
Protein	Signal transducer and activator of transcription 1 (STA-1)	UniProt
Gene / ORF	<i>sta-1</i> / Y51H4A.17	UniProt (Name from PMID: 16401427)
Organism	<i>Caenorhabditis elegans</i>	UniProt
Length	706 aa; isoforms a/b/c	UniProt
Family	STAT transcription factor family	UniProt / InterPro
Key domains	p53-like DNA-binding (IPR008967), SH2 (IPR000980), STAT (IPR001217), STAT linker (IPR048988)	InterPro
Subcellular location	Cytoplasm and Nucleus	UniProt

The gene symbol, organism, family, and domain architecture are all internally consistent and match the primary literature (PMID: 16873887, PMID: 16401427). **The one critical caveat is paralog ambiguity within the same organism:** *C. elegans* has a second STAT-like factor, **STA-2**, with a largely distinct biology (epidermal innate immunity). Literature referring to "the worm STAT" must be read carefully to assign findings to the correct paralog. This report attributes STA-2-specific findings to STA-2 and reserves STA-1 claims for the *sta-1* gene product.

Key Findings

Finding 1 — STA-1 is a bona fide STAT transcription factor that binds a conserved DNA element but lacks the N-terminal oligomerization domain

Wang & Levy (2006), in *C. elegans STAT: evolution of a regulatory switch* (PMID: 16873887), structurally and functionally characterized STA-1 as the *C. elegans* STAT ortholog. The protein contains the canonical STAT architecture — coiled-coil, p53-like DNA-binding domain, linker, and SH2 domain — and recognizes a GAS-like *cis* DNA element that is conserved across phylogeny. In the authors' words, "*this protein, termed STA-1, is structurally and functionally related to other vertebrate and invertebrate STAT proteins, recognizing a cis DNA element conserved through phylogeny.*"

The distinctive structural departure is that "*STA-1 lacks the conserved amino-terminal oligomerization domain found in vertebrate and other invertebrate STAT proteins.*" In mammalian STATs, this N-terminal domain mediates tetramerization on tandem DNA sites and stabilizes cooperative, high-avidity binding. Its absence in STA-1 — paralleled by its absence in a distant nematode STAT and in the *Dictyostelium* STAT — predicts altered DNA-binding cooperativity and is consistent with a regulatory "switch" in STAT evolution. This domain architecture (confirmed by InterPro: IPR001217 STAT, IPR000980 SH2, IPR008967 p53-like DNA-binding) establishes the molecular toolkit by which STA-1 executes sequence-specific transcriptional control.

Finding 2 — STA-1 represses dauer formation, cooperating non-canonically with a DAF-7/TGF- β branch and requiring tyrosine phosphorylation

Wang & Levy (2006), *C. elegans* STAT cooperates with DAF-7/TGF- β signaling to repress dauer formation (PMID: 16401427), provided the first in vivo functional assignment for STA-1. Activated STA-1 "accumulated in the nuclei of five head neuron pairs, three of which are amphid neurons involved in dauer formation," localizing its action to the sensory neurons that integrate environmental cues governing the dauer (diapause) decision.

Genetically, "*sta-1* mutants showed a synthetic dauer phenotype with selected TGF-beta mutations," demonstrating that STA-1 cooperates with TGF- β signaling to suppress dauer entry. Crucially, the rescue experiment pinned down the biochemical requirement: "*sta-1* deficiency was complemented by reconstitution with wild-type protein, but not with a tyrosine mutant," establishing that a phosphorylatable tyrosine — the canonical STAT activation residue engaged by the SH2 domain during dimerization — is required for function.

The pathway logic is non-canonical. STA-1 functions in the **absence** of DAF-7, DAF-4, and DAF-14 but **requires** DAF-1 and DAF-8, and its own expression is induced by TGF- β signaling in a **DAF-3-dependent** manner, constituting a negative-feedback loop in which TGF- β targets DAF-3 to keep neuronal STA-1 expression in check. This places STA-1 as a repressive node embedded within, but partially independent of, the classical dauer TGF- β cascade.

Finding 3 — STA-1 acts as a transcriptional REPRESSOR in antiviral immunity, opposite to activating mammalian STATs

Tanguy et al. (2017), *An Alternative STAT Signaling Pathway Acts in Viral Immunity in C. elegans* (PMID: 28874466), used Orsay virus infection combined with gene-expression profiling and chromatin immunoprecipitation to show that "*the C. elegans* STAT homolog STA-1 orchestrates antiviral immunity. Intriguingly, mutants lacking STA-1 are less permissive to antiviral infection." Because deleting STA-1 makes animals *more* resistant, STA-1 must normally **suppress** a constitutively poised antiviral response. The authors state plainly that "*in contrast to the mammalian pathway, STA-1 acts mostly as a transcriptional repressor.*"

This study also identified a new signaling partner: "*we identify the kinase SID-3 as a new component of the response to infection, which, along with STA-1, participates in the transcriptional regulatory network of the immune response*" — and notably this activity is independent of the RNAi machinery.

Batachari et al. (2026), *Viral infection drives cell-intrinsic re-localization of the C. elegans STA-1* (PMID: 42427644), confirmed and extended the repressor model: "*STA-1 overexpression causes increased susceptibility to viral infection, in a manner dependent on conserved residues important for DNA binding, nuclear localization and phosphorylation.*" The dose-dependence (more STA-1 → more susceptibility) and the dependence on DNA-binding, nuclear-localization, and phosphorylation residues demonstrate that the repressive effect requires STA-1 to function as an intact, activatable transcription factor in the nucleus.

Finding 4 — STA-1 acts cell-intrinsically and relocates from the nucleus to DRH-1-associated cytoplasmic puncta upon infection; it is structurally STAT5-like

Batachari et al. (2026, PMID: 42427644) resolved the subcellular dynamics underlying the repressive switch. "*C. elegans STA-1 protein disappears from nuclei of cells infected with the natural viral pathogen, Orsay virus, but remains nuclear in uninfected cells, indicating a cell-intrinsic site of action.*" The loss of STA-1 from infected-cell nuclei provides a direct mechanistic account for de-repression: with the repressor removed, defense genes are released.

Infection does not simply degrade STA-1; it redistributes it. "*During viral infection, STA-1 forms cytoplasmic puncta that interact with the RNA viral sensor DRH-1,*" the RIG-I-like helicase that senses viral RNA. This interaction suggests DRH-1 restrains the immune-repressive factor by sequestering it in the cytoplasm, coupling viral detection directly to the subcellular relocation of a transcriptional repressor.

Structurally, "*structural predictions indicate that STA-1 is most similar to STAT5 proteins in mammals, which have known immune-repressive roles.*" This structural classification rationalizes the otherwise paradoxical repressive behavior — STA-1's closest mammalian analog is itself associated with transcriptional repression rather than interferon-type activation. Consistent with broad physiological reach, Wang & Levy (PMID: 16873887) reported widespread STA-1 expression across multiple worm tissues and showed that an activated mutant STA-1 can accumulate in the nucleus even without functional DNA-binding/coiled-coil domains — decoupling nuclear import from DNA engagement.

Finding 5 — STA-1 is distinct from its paralog STA-2; STA-1 is a 706-aa SH2-domain protein with broad expression and dual cytoplasmic/nuclear localization

The UniProt record for Q9NAD6 specifies a 706-aa protein with an SH2 domain at residues ~477–574, three isoforms (a/b/c), and a subcellular location in both **cytoplasm and nucleus**. Tissue specificity spans the adult/larval pharynx, head ganglia, tail ganglia, ventral nerve cord, and body-wall muscles — a broad expression pattern consistent with pleiotropic regulatory roles. The annotated molecular function combines **signal transduction and transcriptional activation**, with the activated STAT repressing dauer formation and its neuronal expression held in check by TGF- β acting through DAF-3.

Critically, STA-1 must not be conflated with **STA-2**, a separate gene product and the STAT-like factor responsible for **epidermal antimicrobial-peptide (AMP) responses**. Dierking et al. (2011, [PMID: 21575913](#)) showed that "*the STAT transcription factor-like protein STA-2 as a direct physical interactor of SNF-12 and show that the two proteins function together to regulate AMP gene expression in the epidermis*," partnering with the SLC6-family transporter SNF-12. Zhang et al. (2015, [PMID: 25692704](#)) demonstrated that structural damage causes "*detachment of STA-2 molecules from hemidesmosomes and transcription of AMPs*," tying STA-2 to the apical hemidesmosome and a p38 MAPK pathway responding to the fungus *Drechmeria coniospora*. These are functions of **STA-2**, **not STA-1**, and are included here to delineate the boundary precisely.

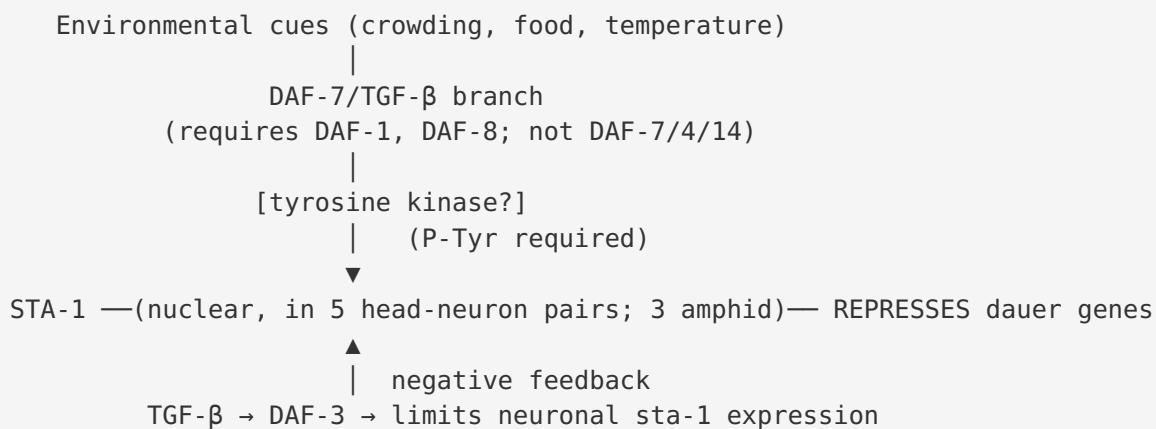
Finding 6 — STA-1 is activated by non-JAK tyrosine phosphorylation, with SID-3 as a tyrosine-kinase partner

C. elegans lacks both JAK kinases and the interferon system (Tanguy et al. 2017, [PMID: 28874466](#)), yet STA-1 function strictly requires a phosphorylatable tyrosine — a tyrosine-mutant protein fails to rescue the *sta-1* null (Wang & Levy 2006, [PMID: 16401427](#)). Together these facts imply that an **alternative (non-JAK) tyrosine kinase** activates STA-1. The best genetic candidate is **SID-3**, identified alongside STA-1 in the antiviral transcriptional network: "*we identify the kinase SID-3 as a new component of the response to infection, which, along with STA-1, participates in the transcriptional regulatory network of the immune response*" ([PMID: 28874466](#)). UniProt Q10925 confirms SID-3 is a tyrosine-protein kinase — the worm ACK1/TNK2 ortholog, otherwise known for mediating systemic RNAi through dsRNA import/export. Whether SID-3 directly phosphorylates STA-1 or acts more indirectly in the network remains to be established biochemically, but the genetic association provides a concrete lead for the identity of the activating kinase.

Mechanistic Model / Interpretation

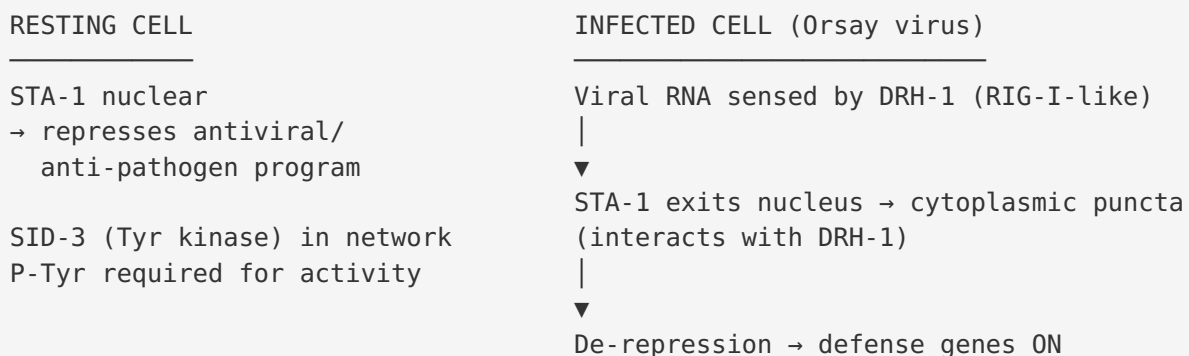
STA-1 can be understood as a **repressive STAT "brake"** that is poised on defense and developmental programs and is released by specific upstream signals. Two parallel contexts share this logic.

Context A — Dauer decision (neurons):



In the sensory neurons that read environmental state, activated (tyrosine-phosphorylated) STA-1 accumulates in nuclei and represses the dauer program, cooperating with a specific DAF-1/DAF-8 branch of TGF-β signaling. A DAF-3-dependent feedback loop keeps STA-1 levels tuned. Loss of STA-1 sensitizes animals to inappropriate dauer entry only in combination with TGF-β mutations (a synthetic phenotype), indicating STA-1 provides a parallel, reinforcing repressive input rather than being the sole switch.

Context B — Antiviral defense (intestine):



Here STA-1 is a **negative regulator**: in uninfected intestinal cells it is nuclear and suppresses a constitutively available antiviral response. Infection, sensed by DRH-1, triggers STA-1's exit from the nucleus into DRH-1-associated cytoplasmic puncta, lifting repression and allowing defense (including later-induced anti-pathogen genes) to proceed. The requirement for DNA-binding, nuclear-localization, and phosphorylation residues — and the dose-dependent increase in viral susceptibility upon STA-1 overexpression — all confirm that the repressive output depends on STA-1 functioning as an intact nuclear transcription factor.

Unifying features:

Feature	STA-1 (worm)	Classical mammalian STAT (e.g., STAT1)
Primary output	Repression	Activation
Upstream kinase	Non-JAK (SID-3 candidate)	JAK family
Interferon system	Absent in <i>C. elegans</i>	Central
N-terminal oligomerization domain	Absent	Present
Closest mammalian analog	STAT5 (immune-repressive)	—
Activation residue	Phospho-tyrosine (required)	Phospho-tyrosine
Localization logic	Nuclear when repressing; cytoplasmic puncta when disabled	Cytoplasmic→nuclear upon activation

The recurring theme is an **inversion of the canonical STAT paradigm**: STA-1 is a phospho-tyrosine-dependent STAT whose default active state *silences* programs, and whose removal from the nucleus (developmentally via feedback, or acutely via DRH-1 during infection) *de-represses* them. Its structural kinship to STAT5 and its missing oligomerization domain provide plausible molecular bases for this repressive wiring.

Evidence Base

PMID	Title (abbrev.)	Contribution	Relationship to findings
16873887	<i>C. elegans STAT: evolution of a regulatory switch</i>	Structural/functional characterization; conserved <i>cis</i> element; missing N-terminal domain; broad expression	Supports F1, F4
16401427	<i>C. elegans STAT cooperates with DAF-7/ TGF-β to repress dauer formation</i>	Neuronal nuclear localization; synthetic dauer with TGF- β ; tyrosine requirement; DAF-1/DAF-8/DAF-3 logic	Supports F2, F6
28874466	<i>An Alternative STAT Signaling Pathway Acts in Viral Immunity</i>	STA-1 is a repressor; <i>sta-1</i> null is more resistant; SID-3 kinase identified; no JAK/interferon	Supports F3, F6
42427644	<i>Viral infection drives cell-intrinsic re-localization of the C. elegans STAT</i>	Nuclear exit on infection; DRH-1-associated cytoplasmic puncta; STAT5-like; overexpression $\rightarrow$ susceptibility	Supports F3, F4
21575913	<i>Unusual regulation of a STAT by an SLC6 transporter</i>	STA-2–SNF-12 partnership in epidermal AMP regulation	Delineates STA-2 (not STA-1); supports F5
25692704	<i>Structural damage causes release of STA-2</i>	STA-2 at hemidesmosomes; release drives AMP transcription	Delineates STA-2; supports F5
34166401	<i>Antagonistic fungal enterotoxins...</i>	Enterotoxins modulate STA-2 nuclear levels / AMP expression	Context for STA-2 paralog (not STA-1)
33259791	<i>Innate Immunity Promotes Sleep through Epidermal AMPs</i>	STA-2/SNF-12 AMP axis links immunity to sleep	Context for STA-2 paralog (not STA-1)
29405821	<i>Modulatory upregulation of an insulin peptide gene...</i>	STA-2/STAT drives epidermal <i>ins-11</i> via p38 MAPK	Context for STA-2 paralog (not STA-1)

Consistency of the evidence. Four studies bear directly on STA-1 (PMID: 16873887, 16401427, 28874466, 42427644). They are mutually reinforcing: two independent physiological contexts (dauer, antiviral) converge on the same molecular picture — a phospho-tyrosine-dependent STAT that acts as a repressor and whose nuclear presence correlates with silencing. The remaining five papers concern STA-2 and were used to firewall STA-1's identity from its paralog, preventing misattribution.

Limitations and Knowledge Gaps

- 1. Identity of the activating tyrosine kinase is unresolved.** A phosphorylatable tyrosine is required for STA-1 activity (PMID: 16401427), and SID-3 is a genetic partner in the antiviral network (PMID: 28874466), but **direct biochemical phosphorylation of STA-1 by SID-3 has not been demonstrated**. The kinase acting in the neuronal/dauer context is likewise unknown.
 - 2. Direct target genes and DNA occupancy are incompletely mapped.** Although a conserved GAS-like *cis* element is recognized (PMID: 16873887) and ChIP was used in the antiviral study (PMID: 28874466), a comprehensive, condition-resolved map of direct STA-1 target promoters in neurons versus intestine is lacking.
 - 3. Mechanism of repression at the molecular level.** How STA-1 represses — whether by recruiting corepressors, competing with activators, or occluding activating factor binding — remains undefined. Its STAT5-like structure and missing oligomerization domain are suggestive but not mechanistically proven.
 - 4. Nature of the DRH-1 interaction.** Whether DRH-1 directly binds STA-1 or co-localizes within a larger complex, and whether sequestration is the cause or consequence of nuclear exit, needs biochemical and temporal dissection (PMID: 42427644).
 - 5. Isoform-specific functions.** Three isoforms (a/b/c) are annotated (UniProt Q9NAD6), but their distinct expression and functional roles are uncharacterized.
 - 6. Reliance on abstract-level evidence.** Several key mechanistic claims rest on abstracts rather than full-text extraction of quantitative effect sizes; precise phenotypic penetrance, fold-changes, and statistics were not all independently verified here.
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Proposed Follow-up Experiments / Actions

1. **Test SID-3 as the direct STA-1 kinase.** Perform in vitro kinase assays with recombinant SID-3 and STA-1, and map the phosphorylated tyrosine by mass spectrometry; validate in vivo with phospho-specific antibodies in *sid-3* mutant vs. wild-type backgrounds, in both infected intestine and dauer-relevant neurons.
 2. **Map STA-1 genome-wide occupancy by context.** Perform ChIP-seq (or CUT&RUN) for STA-1 in uninfected vs. Orsay-virus-infected intestine and in dauer-permissive conditions, to define direct repressed targets and test for the conserved GAS-like motif at those sites.
 3. **Define the repression mechanism.** Use IP-mass spectrometry to identify STA-1 corepressor partners; test whether the missing N-terminal oligomerization domain (versus a chimeric STA-1 with a grafted domain) alters repressive output and target selection.
 4. **Dissect the DRH-1–STA-1 switch.** Use live imaging with tagged STA-1 and DRH-1 to time nuclear exit relative to infection; test whether *drh-1* loss abolishes STA-1 puncta and prevents de-repression, and whether forced nuclear retention of STA-1 blocks the antiviral response.
 5. **Identify the neuronal dauer kinase.** Conduct a candidate/RNAi screen of *C. elegans* tyrosine kinases for loss of STA-1 tyrosine phosphorylation and for *sta-1*-like synthetic dauer phenotypes with TGF- β mutants.
 6. **Resolve isoform roles.** Generate isoform-specific reporters and CRISPR deletions to assign a/b/c isoforms to the neuronal-dauer vs. intestinal-antiviral functions.
 7. **Structural validation.** Experimentally determine or refine the STA-1 structure (or validate an AlphaFold model) to test the predicted STAT5 kinship and examine the DNA-binding and SH2 interfaces that underlie its repressive, non-oligomerizing behavior.
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Conclusion

STA-1 is the canonical *C. elegans* STAT: a 706-aa, SH2-domain transcription factor that binds a conserved STAT *cis*-element in a tyrosine-phosphorylation-dependent manner but lacks the N-terminal oligomerization domain of other STATs. Unlike activating mammalian STATs, and operating without any JAK or interferon system, STA-1 functions principally as a **nuclear transcriptional repressor**. It represses dauer formation in head/amphid neurons in

cooperation with a non-canonical DAF-1/DAF-8 branch of DAF-7/TGF- β signaling (with DAF-3-dependent feedback), and it acts cell-intrinsically in the intestine as a negative regulator of the antiviral/anti-pathogen program — being exported from the nucleus into DRH-1-associated cytoplasmic puncta upon Orsay virus infection to de-repress defense. Its closest mammalian relative is the immune-repressive STAT5, and the ACK1/TNK2-family kinase SID-3 is its leading candidate activating kinase. The epidermal antimicrobial-peptide STAT is the distinct paralog **STA-2**, which should not be conflated with STA-1.

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